

Marketing-roi-analysis using Simple Linear Regression

1. Import Libraries

```
In [1]: # Data manipulation
import pandas as pd
import numpy as np

# Visualization
import matplotlib.pyplot as plt
import seaborn as sns

# Statistical modeling
import statsmodels.api as sm

# Model evaluation
from sklearn.metrics import mean_squared_error
```

2. Load the Dataset

```
In [2]: # Load dataset
df = pd.read_csv("marketing_and_sales_data_evaluate_lr.csv")

# Display first 5 rows
print(df.head())
```

	TV	Radio	Social_Media	Sales
0	16.0	6.566231	2.907983	54.732757
1	13.0	9.237765	2.409567	46.677897
2	41.0	15.886446	2.913410	150.177829
3	83.0	30.020028	6.922304	298.246340
4	15.0	8.437408	1.405998	56.594181

3. Explore the Dataset

a. Check Dataset Information

```
In [3]: print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
#   Column          Non-Null Count  Dtype
---  -
0   TV               4562 non-null   float64
1   Radio            4568 non-null   float64
2   Social_Media     4566 non-null   float64
3   Sales            4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
None
```

b. Checking the shape

```
In [4]: df.shape
```

```
Out[4]: (4572, 4)
```

c. Checking missing value

```
In [5]: df.isna().sum()
```

```
Out[5]: TV                10
Radio                  4
Social_Media          6
Sales                  6
dtype: int64
```

d. Removing missing values

```
In [6]: df = df.dropna()
```

e. Statistical Summary

```
In [7]: print(df.describe())
```

	TV	Radio	Social_Media	Sales
count	4546.000000	4546.000000	4546.000000	4546.000000
mean	54.062912	18.157533	3.323473	192.413332
std	26.104942	9.663260	2.211254	93.019873
min	10.000000	0.000684	0.000031	31.199409
25%	32.000000	10.555355	1.530822	112.434612
50%	53.000000	17.859513	3.055565	188.963678
75%	77.000000	25.640603	4.804919	272.324236
max	100.000000	48.871161	13.981662	364.079751

4. Exploratory Data Analysis (EDA)

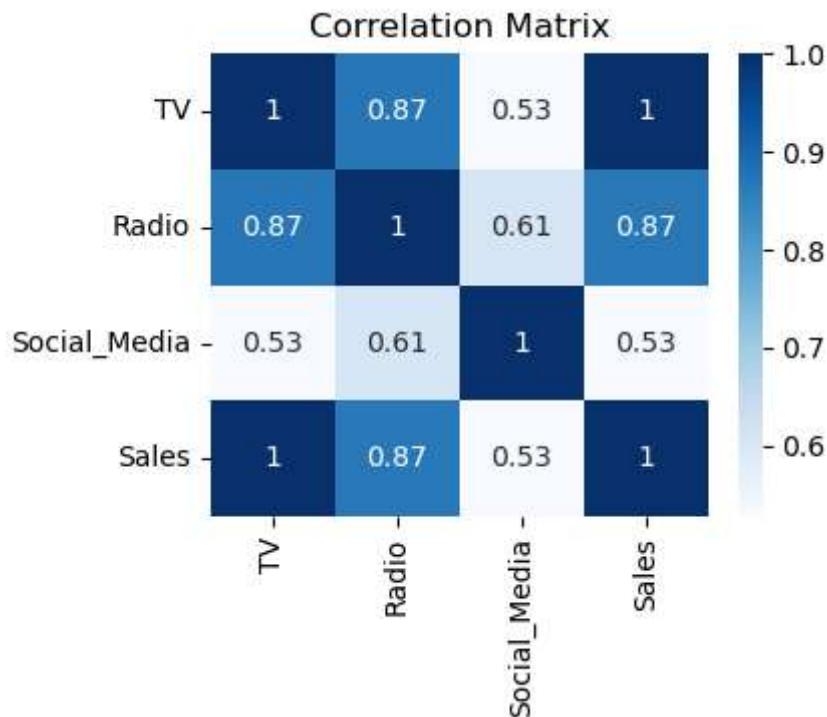
a. Correlation Heatmap

```
In [8]: plt.figure(figsize=(4,3))

sns.heatmap(df.corr(), annot=True, cmap='Blues')

plt.title("Correlation Matrix")
plt.show()

print("\nThis helps identify which advertising channel has the strongest relationsh
```



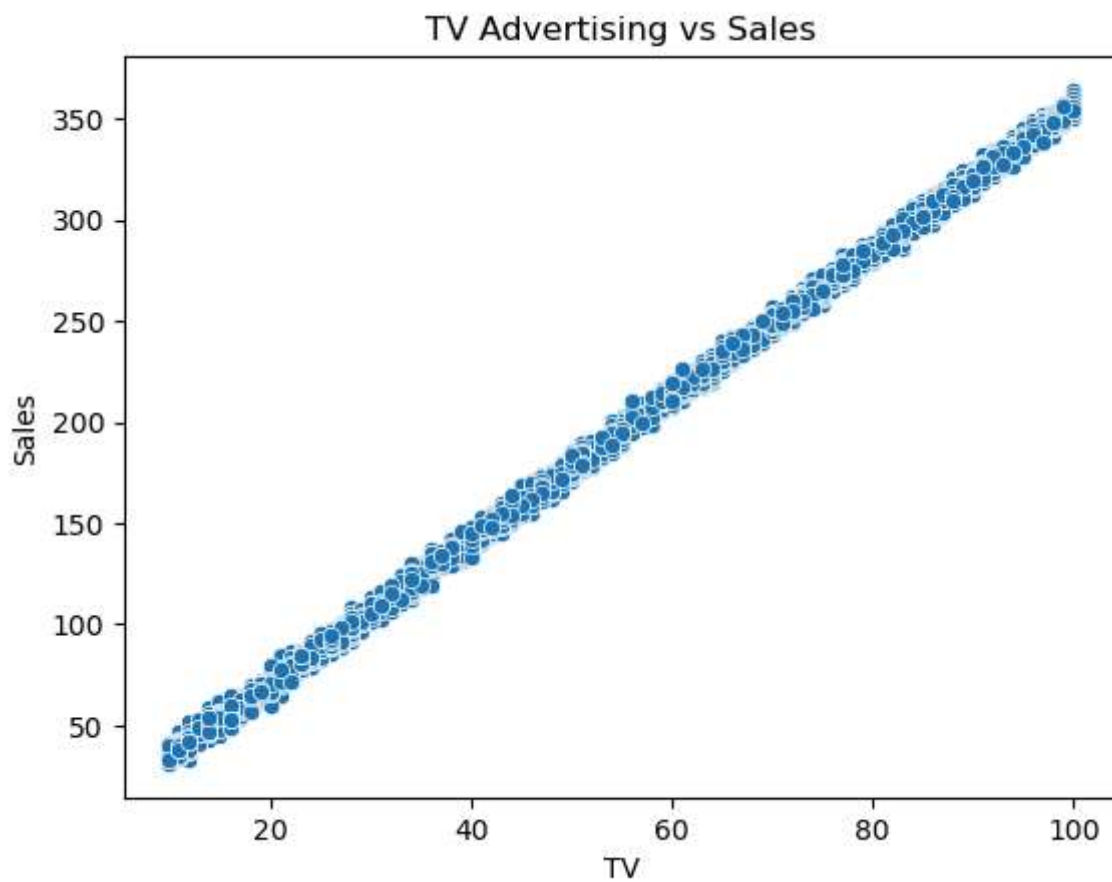
This helps identify which advertising channel has the strongest relationship with Sales.

5. Visualize Relationships

a. TV vs Sales

```
In [9]: sns.scatterplot(x='TV', y='Sales', data=df)

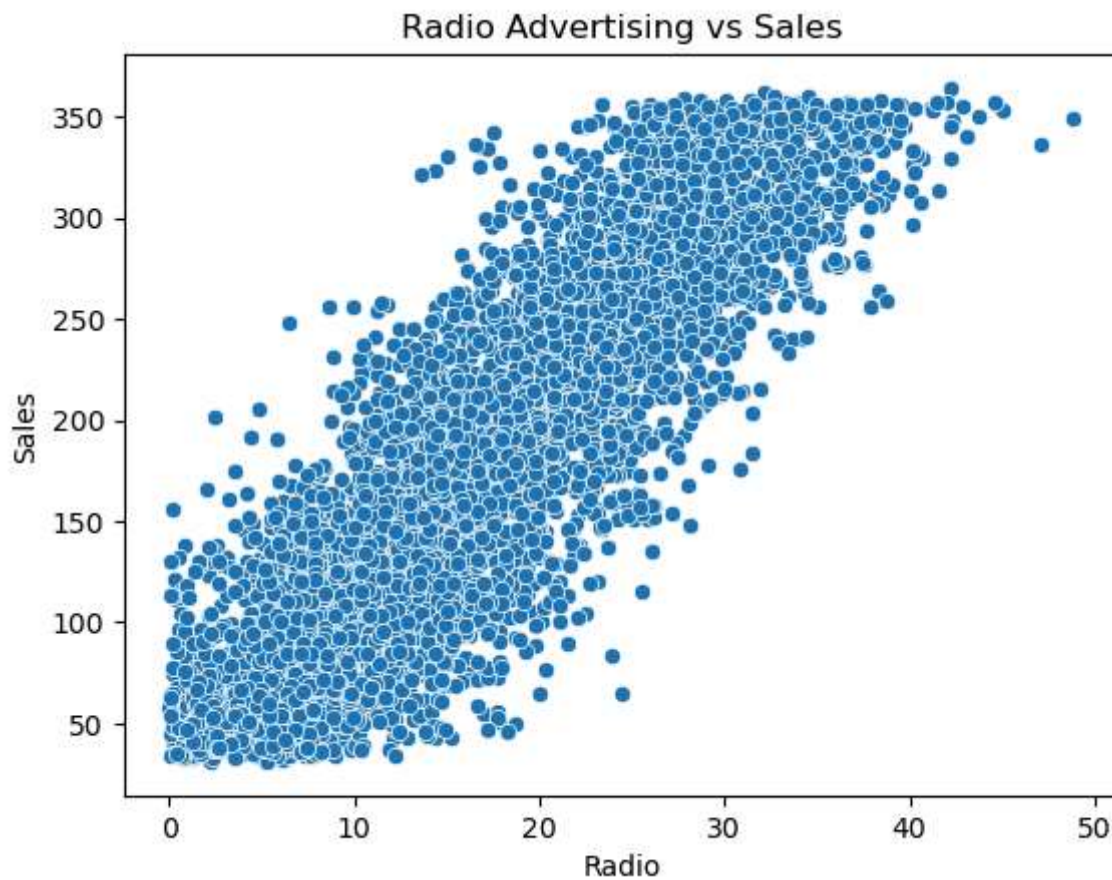
plt.title("TV Advertising vs Sales")
plt.show()
```



b. Radio vs Sales

```
In [10]: sns.scatterplot(x='Radio', y='Sales', data=df)

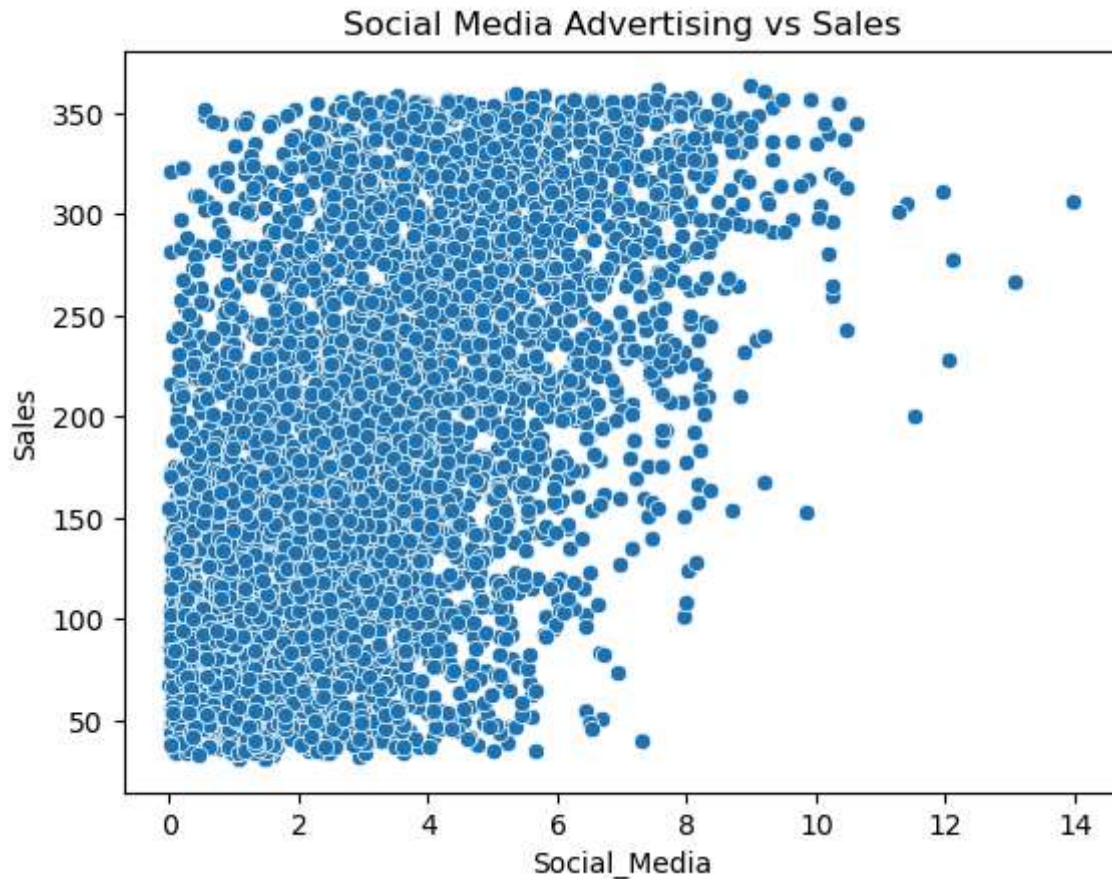
plt.title("Radio Advertising vs Sales")
plt.show()
```



c. Social Media vs Sales

```
In [11]: sns.scatterplot(x='Social_Media', y='Sales', data=df)

plt.title("Social Media Advertising vs Sales")
plt.show()
```



6. Select the Best Independent Variable

```
In [12]: print("Variable      Correlation with Sales")
print("-----")
print(" TV              1.00      ")
print(" Radio             0.87      ")
print(" Social Media       0.53      ")

print("\nTV is the best predictor because it has the highest positive correlation w
```

Variable	Correlation with Sales
TV	1.00
Radio	0.87
Social Media	0.53

TV is the best predictor because it has the highest positive correlation with Sales.

7. Building the Simple Linear Regression Model

Using:

Independent Variable (X) → TV

Dependent Variable (y) → Sales

```
In [13]: X = df['TV']
         y = df['Sales']
```

a. Add Constant

```
In [14]: X = sm.add_constant(X)
```

b. Fit the Model

```
In [15]: model = sm.OLS(y, X).fit()
```

c. Display Results

```
In [16]: print(model.summary())
```

```

                        OLS Regression Results
=====
Dep. Variable:          Sales    R-squared:                0.999
Model:                  OLS      Adj. R-squared:            0.999
Method:                 Least Squares    F-statistic:          4.517e+06
Date:                  Thu, 11 Jun 2026    Prob (F-statistic):      0.00
Time:                  22:17:51    Log-Likelihood:         -11366.
No. Observations:      4546    AIC:                   2.274e+04
Df Residuals:          4544    BIC:                   2.275e+04
Df Model:               1
Covariance Type:       nonrobust
=====
               coef    std err          t      P>|t|      [0.025    0.975]
-----
const        -0.1325     0.101     -1.317     0.188     -0.330     0.065
TV             3.5615     0.002   2125.272     0.000      3.558     3.565
=====
Omnibus:                 0.052    Durbin-Watson:           1.998
Prob(Omnibus):            0.974    Jarque-Bera (JB):         0.031
Skew:                    -0.001    Prob(JB):                 0.985
Kurtosis:                 3.012    Cond. No.                  138.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

8. Predicted Values

```
In [17]: predictions = model.predict(X)

         print(predictions.head())
```



```
0    56.851733
1    46.167191
2   145.889585
3   295.473177
4    53.290219
dtype: float64
```

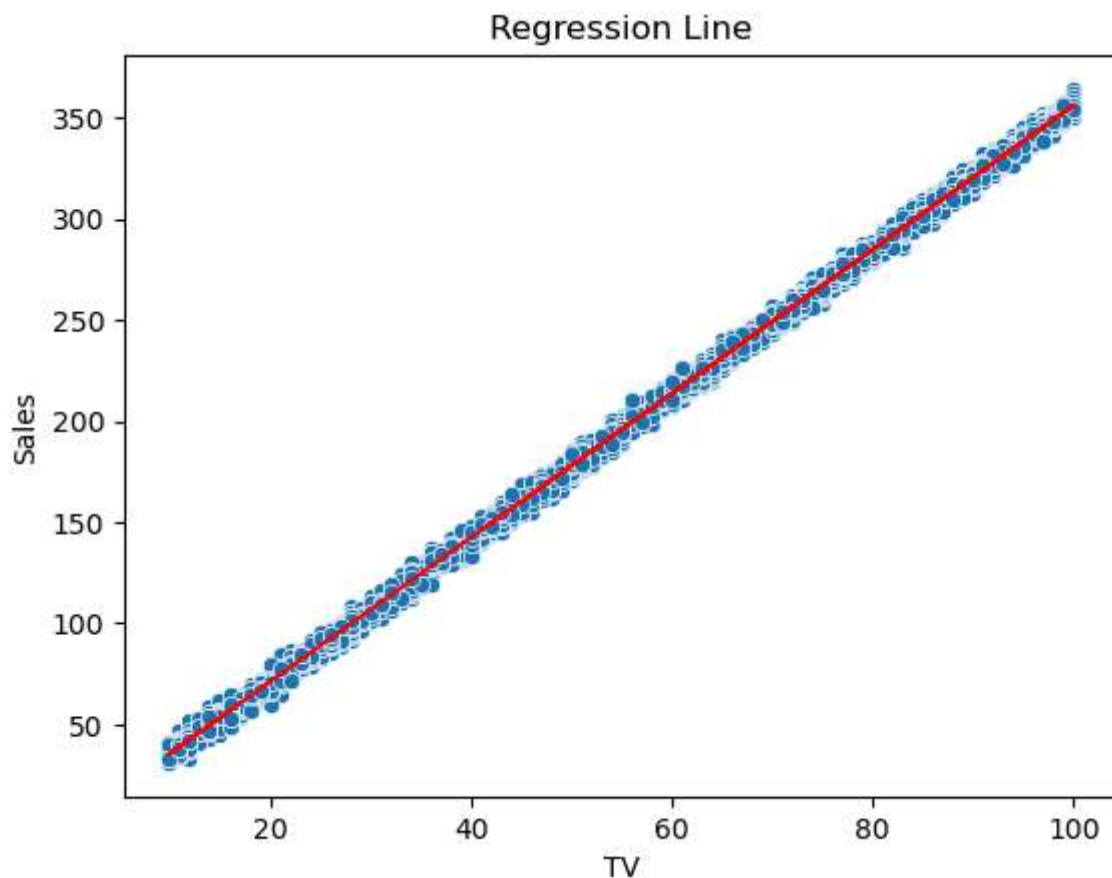
10. Diagnostic Plots

These help verify regression assumptions.

a. Linearity Check

```
In [18]: sns.scatterplot(x=df['TV'], y=y)
plt.plot(df['TV'], predictions, color='red')

plt.title("Regression Line")
plt.show()
print("\nA straight-line trend indicates linearity")
```



A straight-line trend indicates linearity

b. Residual Plot (Homoscedasticity)

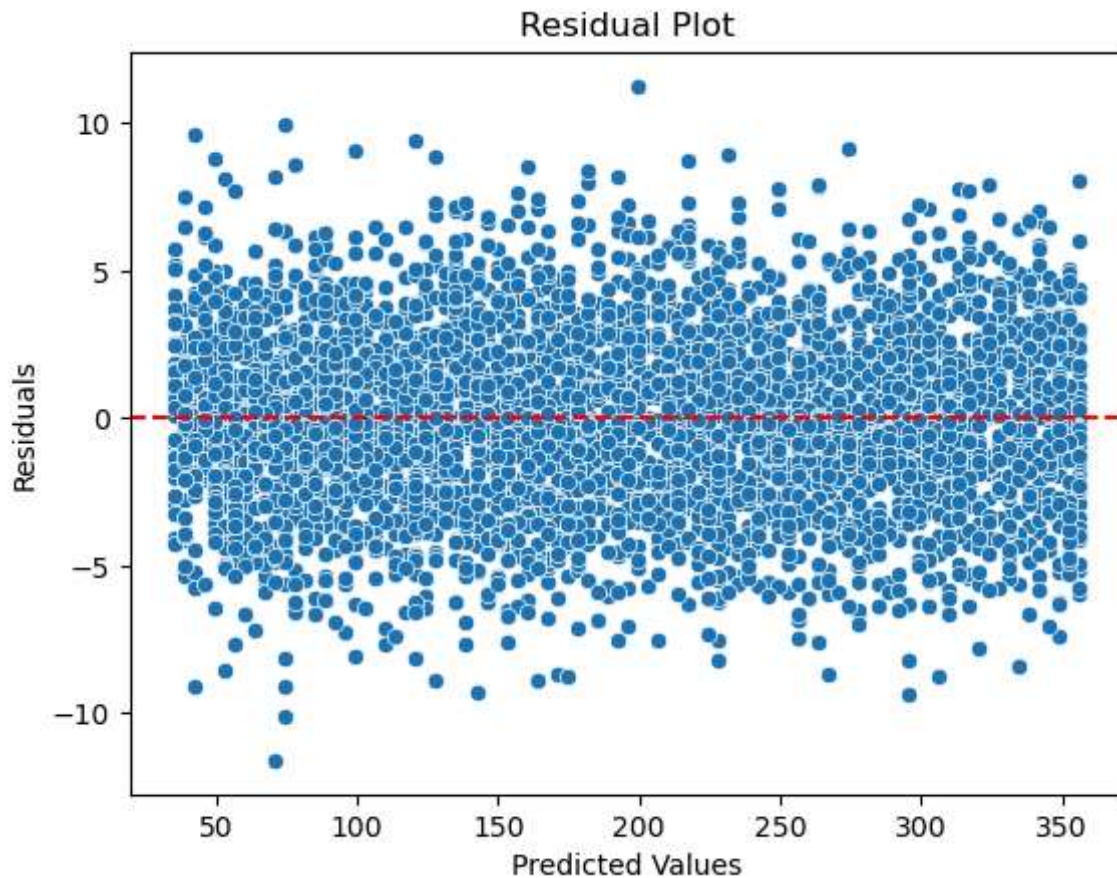
```
In [19]: residuals = y - predictions
```



```
sns.scatterplot(x=predictions, y=residuals)

plt.axhline(y=0, color='red', linestyle='--')

plt.title("Residual Plot")
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.show()
print("\nResiduals should appear randomly distributed around zero.")
```

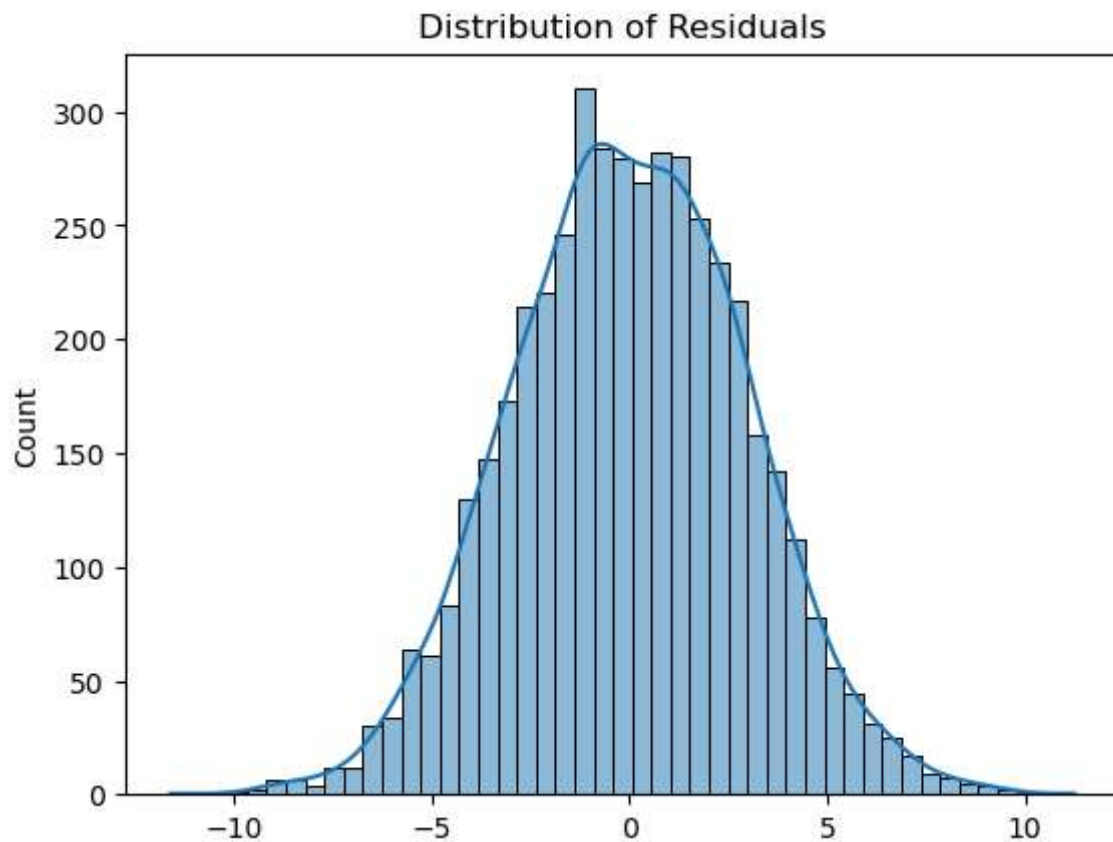


Residuals should appear randomly distributed around zero.

c. Normality Check

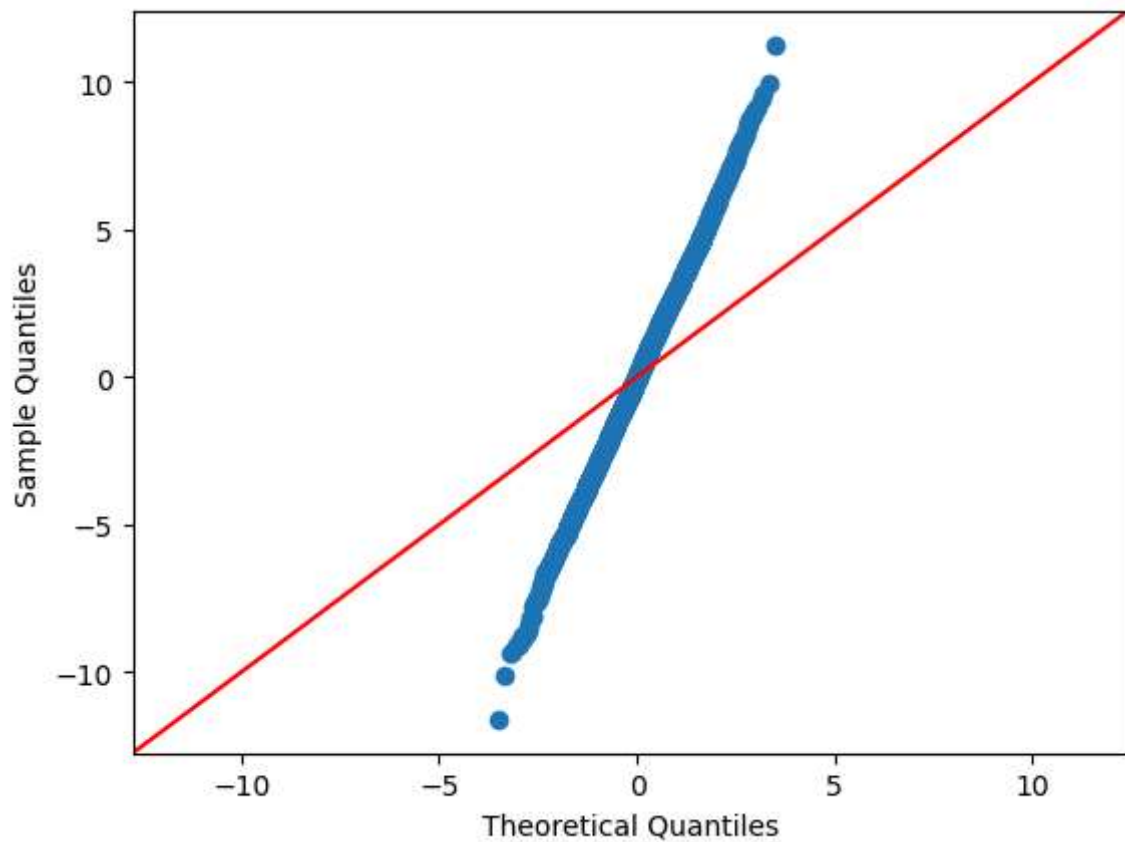
```
In [20]: sns.histplot(residuals, kde=True)

plt.title("Distribution of Residuals")
plt.show()
```



d. Q-Q Plot

```
In [21]: sm.qqplot(residuals, line='45')  
plt.show()
```



e. Histogram of Residuals

11. Recommended Marketing Strategy

Based on the regression analysis: TV advertising has the strongest relationship with Sales. The regression model is statistically significant. The high R-squared value indicates good predictive strength. Diagnostic plots confirm the model assumptions are reasonably satisfied.

12. ROI Recommendation

The company should allocate a larger percentage of its marketing budget toward TV advertising because it produces the strongest measurable impact on sales revenue.